EXPERIMENT – 10

AIM :

To generate a dataset containing at least 10,000 rows, identify and analyze five slow SQL queries using EXPLAIN, optimize them using indexing and query rewriting, and compare query execution time before and after optimization.

SOURCE CODE :

```
-- Use the database selected by the editor account.
-- Creating/selecting databases may be restricted in the sandbox.

-- Create Table
CREATE TABLE Sales (
    SaleID INT PRIMARY KEY,
    CustomerID INT,
    ProductID INT,
    Amount DECIMAL(10,2),
    SaleDate DATE,
    City VARCHAR(50)
);

-- Generate 10000 Records
DELIMITER //

CREATE PROCEDURE GenerateData()
BEGIN
    DECLARE i INT DEFAULT 1;

    WHILE i <= 10000 DO
        INSERT INTO Sales VALUES (
            i,
            i % 1000,
            i % 100,
            (i % 5000) + 100,
            DATE_ADD('2025-01-01',
                INTERVAL (i % 365) DAY),
            CASE
                WHEN i % 3 = 0 THEN 'Delhi'
                WHEN i % 3 = 1 THEN 'Mumbai'
                ELSE 'Kanpur'
            END
        );

        SET i = i + 1;
    END WHILE;
END //
```

```
DELIMITER ;

CALL GenerateData();

-- Verify Number of Records
SELECT COUNT(*) FROM Sales;

-- QUERY 1: Search by CustomerID
-- Before Optimization
EXPLAIN
SELECT * FROM Sales
WHERE CustomerID = 105;

SELECT * FROM Sales
WHERE CustomerID = 105;

-- Optimization
CREATE INDEX idx_customer
ON Sales(CustomerID);

EXPLAIN
SELECT * FROM Sales
WHERE CustomerID = 105;

-- QUERY 2: Search by City
-- Before Optimization
EXPLAIN
SELECT * FROM Sales
WHERE City = 'Delhi';

-- Optimization
CREATE INDEX idx_city
ON Sales(City);

EXPLAIN
SELECT * FROM Sales
WHERE City = 'Delhi';

-- QUERY 3: Search by Date
-- Before Optimization
EXPLAIN
SELECT * FROM Sales
WHERE SaleDate = '2025-03-01';

-- Optimization
CREATE INDEX idx_date
ON Sales(SaleDate);
```

```
EXPLAIN
SELECT * FROM Sales
WHERE SaleDate = '2025-03-01';

-- QUERY 4: Search by Amount
-- Before Optimization
EXPLAIN
SELECT * FROM Sales
WHERE Amount > 3000;

-- Optimization
CREATE INDEX idx_amount
ON Sales(Amount);

EXPLAIN
SELECT * FROM Sales
WHERE Amount > 3000;

-- QUERY 5: Search by Customer and Date
-- Before Optimization
EXPLAIN
SELECT * FROM Sales
WHERE CustomerID = 105
AND SaleDate = '2025-03-01';

-- Optimization
CREATE INDEX idx_customer_date
ON Sales(CustomerID, SaleDate);

EXPLAIN
SELECT * FROM Sales
WHERE CustomerID = 105
AND SaleDate = '2025-03-01';

-- Query Rewriting Example
-- Original Query
SELECT * FROM Sales
WHERE YEAR(SaleDate) = 2025;

-- Optimized Query
SELECT * FROM Sales
WHERE SaleDate >= '2025-01-01'
AND SaleDate < '2026-01-01';

-- Compare Query Execution Time
-- Execute before and after optimization
```

-- Use EXPLAIN ANALYZE in MySQL 8.0.18+

```
EXPLAIN ANALYZE
SELECT * FROM Sales
WHERE CustomerID = 105;

EXPLAIN ANALYZE
SELECT * FROM Sales
WHERE CustomerID = 105
AND SaleDate = '2025-03-01';
```

OUTPUT :

